

YICHANG, China

WMO: 574610

Lat: 30.7337N

Lon: 111.3596E

Elev: 258

StdP: 98.27

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	0.0	-9.7	1.7	4.9	-7.2	2.1	4.7	4.4	8.7	4.1	7.8	1.3	110	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	35.7	26.6	34.5	26.1	33.2	25.5	28.1	33.1	27.4	32.3	26.8	31.4	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	23.1	31.0	26.2	22.3	30.3	25.6	21.5	29.5	92.4	33.3	89.3	32.4	86.4	31.6	33.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.5	4.0	3.4	-2.5	37.9	1.2	1.0	-3.4	38.6	-4.1	39.2	-4.8	39.8	-5.7	40.5
			-3.9	29.3	1.4	1.2	-4.9	30.1	-5.7	30.8	-6.5	31.5	-7.5	32.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	5.3	7.6	12.6	18.1	22.4	25.8	28.3	27.8	23.9	18.7	13.0	7.6
	DBStd	8.53	3.04	3.71	4.02	3.73	2.99	2.73	2.48	2.67	3.03	3.18	3.68	2.92
	HDD10.0	351	148	83	19	1	0	0	0	0	0	0	14	86
	HDD18.3	1471	404	300	183	48	4	0	0	0	1	34	162	333
	CDD10.0	3138	3	16	99	243	384	473	568	553	416	269	104	11
	CDD18.3	1217	0	0	4	40	130	223	310	295	167	45	2	0
	CDH23.3	11568	0	1	38	299	949	2041	3524	3242	1249	213	12	0
	CDH26.7	4702	0	0	7	83	318	787	1601	1444	422	39	1	0
Wind	WSAvg	1.5	1.3	1.5	1.6	1.6	1.5	1.6	1.6	1.6	1.4	1.3	1.3	1.3
Precipitation	PrecAvg	1145	24	41	58	98	141	154	209	173	101	75	52	19
	PrecMax	1600	62	119	116	269	291	253	450	362	235	192	121	47
	PrecMin	868	3	5	18	16	54	66	70	52	6	7	2	0
	PrecStd	174	16	31	28	51	52	53	85	83	63	46	31	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	20.8	27.1	32.0	33.8	35.6	37.5	37.2	35.0	30.3	25.0	18.6
		MCWB	9.3	12.2	17.6	19.8	21.6	24.9	26.8	26.8	25.5	20.4	15.6	10.6
	2%	DB	13.4	17.4	23.7	29.1	31.6	33.9	36.0	35.5	32.5	27.4	22.2	15.8
		MCWB	7.9	10.8	15.5	19.3	21.3	24.5	27.2	26.5	24.1	18.7	15.2	9.6
	5%	DB	11.4	15.1	21.2	26.8	30.1	32.4	34.8	34.2	30.7	25.7	20.2	13.8
		MCWB	6.9	9.9	14.2	18.6	20.9	24.3	27.0	26.1	23.0	18.0	14.4	8.9
	10%	DB	9.8	13.1	19.0	24.5	28.3	30.9	33.3	32.8	29.1	23.9	18.3	12.0
		MCWB	6.1	8.9	13.4	17.7	20.6	23.9	26.6	25.7	22.0	17.8	13.8	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	14.2	19.5	22.9	25.0	27.6	29.2	28.7	27.0	22.6	17.8	12.1
		MCDB	14.3	17.9	24.6	27.4	29.5	32.4	34.4	34.1	32.9	25.7	20.4	14.7
	2%	WB	8.8	11.9	16.8	21.2	23.7	26.7	28.3	28.0	26.0	20.7	16.6	10.7
		MCDB	12.0	15.3	21.0	25.6	27.8	31.4	33.4	32.9	30.3	24.5	19.8	14.1
	5%	WB	7.7	10.8	15.2	19.9	22.6	25.9	27.7	27.3	25.0	19.6	15.6	9.5
		MCDB	10.8	13.6	19.4	24.3	27.1	30.1	32.7	31.9	28.8	23.4	19.0	13.1
	10%	WB	6.6	9.7	14.0	18.5	21.7	25.2	27.1	26.7	23.9	18.7	14.4	8.6
		MCDB	9.2	12.4	18.0	23.1	26.0	28.6	31.9	31.0	27.0	22.1	17.7	11.5
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.3	7.6	8.2	8.1	7.3	7.4	7.4	6.9	6.8	6.5	6.0
		MCDBR	9.4	10.8	11.9	12.8	11.9	10.1	9.8	9.7	9.7	10.3	10.0	10.2
	5% WB	MCWBR	4.9	5.3	5.1	4.7	4.0	3.2	3.1	2.9	2.9	3.5	4.0	5.0
		MCWBR	8.2	8.4	10.4	10.6	8.8	8.0	8.8	8.6	8.0	7.5	8.1	8.5
Clear-Sky Solar Irradiance	taub		0.618	0.676	0.689	0.680	0.645	0.669	0.630	0.667	0.683	0.672	0.650	0.602
		taud	1.577	1.508	1.522	1.583	1.699	1.688	1.802	1.700	1.635	1.590	1.589	1.628
	Ebn at Noon		578	594	633	666	694	675	700	667	630	589	549	562
		Edh at Noon	226	265	278	270	242	244	217	237	245	240	222	206
All-Sky Solar Radiation	RadAvg		1.97	2.28	3.32	3.93	4.14	4.35	4.78	4.44	3.49	2.85	2.32	1.96
	RadStd		0.36	0.52	0.23	0.45	0.51	0.53	0.55	0.46	0.42	0.42	0.35	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	-0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	+0.42	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page